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Temporally compounding energy droughts in European electricity systems with hydropower

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As Europe's renewable energy capacities expand, electricity systems face increased risks of energy droughts—periods of low production coinciding with high demand. We evaluate characteristics of electricity variability due to weather variations by calculating 1,600 years of daily production and demand. Focusing on five European countries—chosen for their energy mix including hydropower—we find that energy droughts result from processes that cause (temporally) compounding impacts in the energy and meteorological system. These can turn what might have been short-term droughts into prolonged high unmet energy demand. For instance, low reservoir inflows in spring quadruple the chance of prolonged energy droughts: reduced snowpack and rainfall lower hydro availability but also dry out subsoils, increasing the chance of heatwaves and therewith extending the energy problems into summer. We identify and quantify three compounding energy/climate conditions and the associated characteristics and risks of multi-year energy droughts, crucial for informing future energy system design.

As Europe's capacity of renewable electricity sources grows, the region's electricity system is becoming increasingly dependent on variations in weather and climate[1,2]. Electric heating and cooling demand are influenced by air temperatures, and the production of renewables depends on the availability of resources (wind speed, irradiance, discharge). With this increased sensitivity to climate variations arises the risk of encountering weather conditions outside of the system's design scope, undermining energy security. Furthermore, due to nonlinear relationships between climate factors and the electricity system, extremely high residual loads—demand minus renewable energy generation—(hereafter referred to as energy droughts) might occur under compounding meteorological conditions that in itself are not considered extreme[3] and vice versa.

These so-called compound events[4] have recently had substantial impact in electricity systems in a number of European regions. During the autumn of 2021, Europe experienced record-low gas storage levels, a situation influenced by several factors. Besides market prices and geopolitical causes, other contributing factors were a preceding cold winter in Northern Europe followed by another colder-than-usual season in spring, compounded by very low annual average wind speeds in parts of northwestern and central Europe[5]. In September 2021, this prolonged period of calm weather conditions even led to record-high electricity prices in the UK energy market[6]. Furthermore, in the beginning of 2022, Spain and Italy saw a dramatic 40% decline in hydroelectricity generation[7,8]. This sharp drop was caused by an exceptionally hot summer and one of the driest years for water resources since record-keeping began[9]. Consequently, some hydropower plants had to cease operations in the midst of a cooling demand peak[10].

Hydropower is Europe's main source of renewable electricity to date—with an installed capacity of 188 GW, it makes up for 36% of the renewable capacity and supplies 17% of the total EU electricity demand[11]. Whereas hydropower is dispatchable and can be adjusted to

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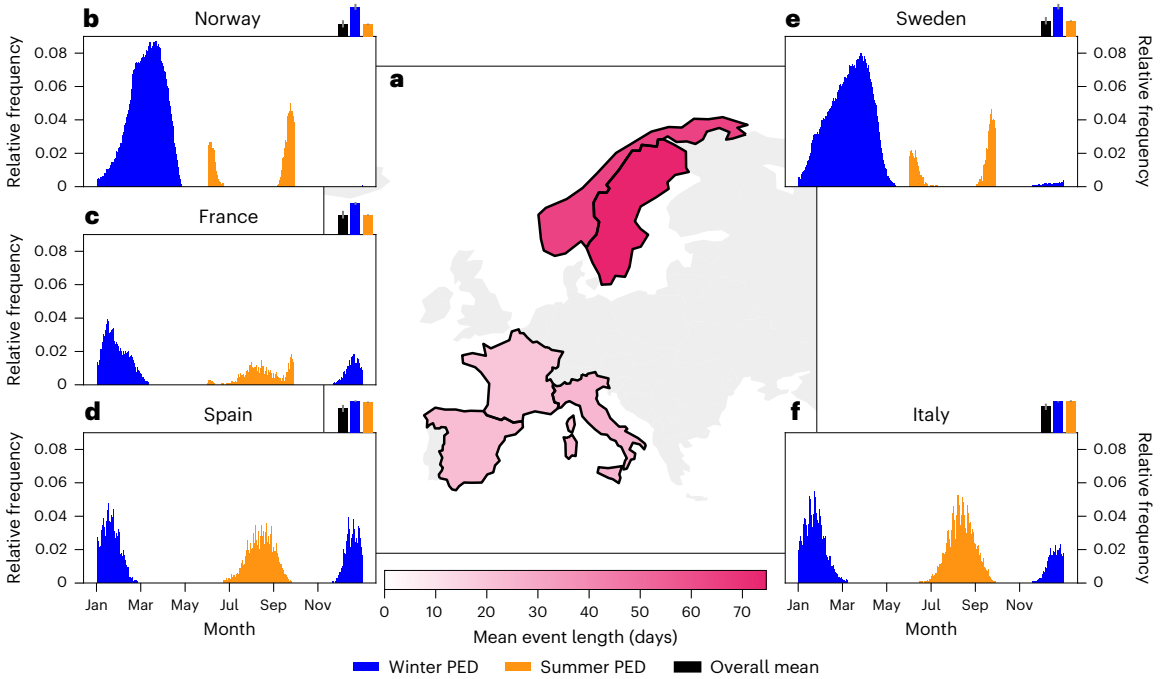

Fig. 1 | PEDs with daily residual loads above the 97th percentile for the once-in-ten-years-or-less PEDs. a, Map of the mean length of the once-in-ten-years-or-less PEDs independent of the season per country. $n = 160$. **b**–**f**, The relative frequency (to the 1,600 years of data) of PED days for each day of the year for the following countries, respectively: Norway (**b**), France (**c**), Spain (**d**), Sweden (**e**) and Italy (**f**). Blue shows winter PEDs (October–May) with the threshold of 97th percentile relative to the residual loads in the winter months. Yellow shows summer PEDs, such as the winter PEDs but for the months June–September. The bar plots on top of the subplots show the mean residual load during the summer and winter PEDs relative to the overall mean residual daily load (in black), with the light grey error bars indicating the standard deviation of the respective residual loads.

meet demand, the amount of water available to produce electricity can vary greatly due to the seasonal variation of the weather and extreme weather events at sub-seasonal timescales. For instance, droughts can reduce water levels in reservoirs, limiting a plant's capacity to generate electricity[12,13].

However, previous studies that modelled energy drought events in Europe's renewable electricity system have focused on wind power and solar photovoltaic (PV) generation[14–20] or only included run-of-river hydropower production[21], overlooking the critical role of reservoir storage and its operational dynamics. One of the reasons hydropower reservoir production is often excluded from studies into the weather sensitivity of electricity systems is that it is characterized by complex operating procedures[22].

Energy system and water management models can be employed to simulate this, but they are typically computationally intensive, leading to reliance on short datasets or estimates of annual energy potential[23]. Recent work applied novel approaches such as the handling of an ensemble of single weather years[24] or synthesizing a set of demand and production profiles based on high-resolution climate data that reflect typical variations to allow for detailed modelling[25]. However, hydropower not only experiences considerable variations from year to year[12]—for instance, changes in the North Atlantic Oscillation index can explain up to 30% of the fluctuations in Norway's hydropower production[26,27]—but might also be susceptible to sustained climate phenomena such as multi-year meteorological or hydrological droughts[12,28,29]. To accurately represent this variability, extended time series data is necessary. The climate and energy modelling community suggest identifying meteorological conditions that have extreme impacts in the electricity system and integrating them into energy system simulations to tackle the problem of handling such large datasets[23].

Here we provide a first-order indication of energy system vulnerabilities to climate variability by identifying meteorological conditions that (temporally) compound to result in high demand and low renewable production using climate model data and a daily renewable electricity production and demand modelling framework[30], designed to be a broad diagnostic tool rather than a predictive one. We employ a large ensemble approach, using 160 sets of ten-year physical climate model simulations (1,600 years)[31], each set representing a different possible sequence of weather under present-day climate conditions (Supplementary Notes 1–3 and Supplementary Figs. 1–7 provide more details of the ensemble). We conduct case studies for five European countries with large hydropower capacities—Norway, France, Italy, Spain and Sweden. Switzerland also has a large hydropower capacity but is omitted following a robustness assessment of our model (Supplementary Note 4 and Supplementary Figs. 8–11) that suggested less reliable results for this country. We evaluate daily residual loads for the full 1,600 years of climate data to: (1) identify and quantify different types of energy drought in the electricity system, (2) analyse temporal compounding in these events and (3) quantify risk ratios of meteorological conditions pre-conditioning energy drought events.

High residual load events in electricity systems with hydropower

We classify high residual load events using two metrics. First, we introduce the concept of 'Energy Drought Windows (EDWs)'[15], which are fixed-duration periods for which we select the highest residual loads from the ensemble. Second, we examine 'Persistent Energy Droughts (PEDs)'[21,32]. These drought events occur when the daily residual load remains consistently above a predefined threshold for consecutive days. Additionally, we differentiate between summer and winter events by defining PED thresholds relative to the respective seasonal residual loads. We examine the top 160 EDWs and PEDs, hereafter referred to as once-in-ten-years-or-less events.

In Norway and Sweden, countries with relatively large hydropower storage capacities characterized by a strong seasonal cycle, PEDs can last more than 150 days (Fig. 1a). These countries can use

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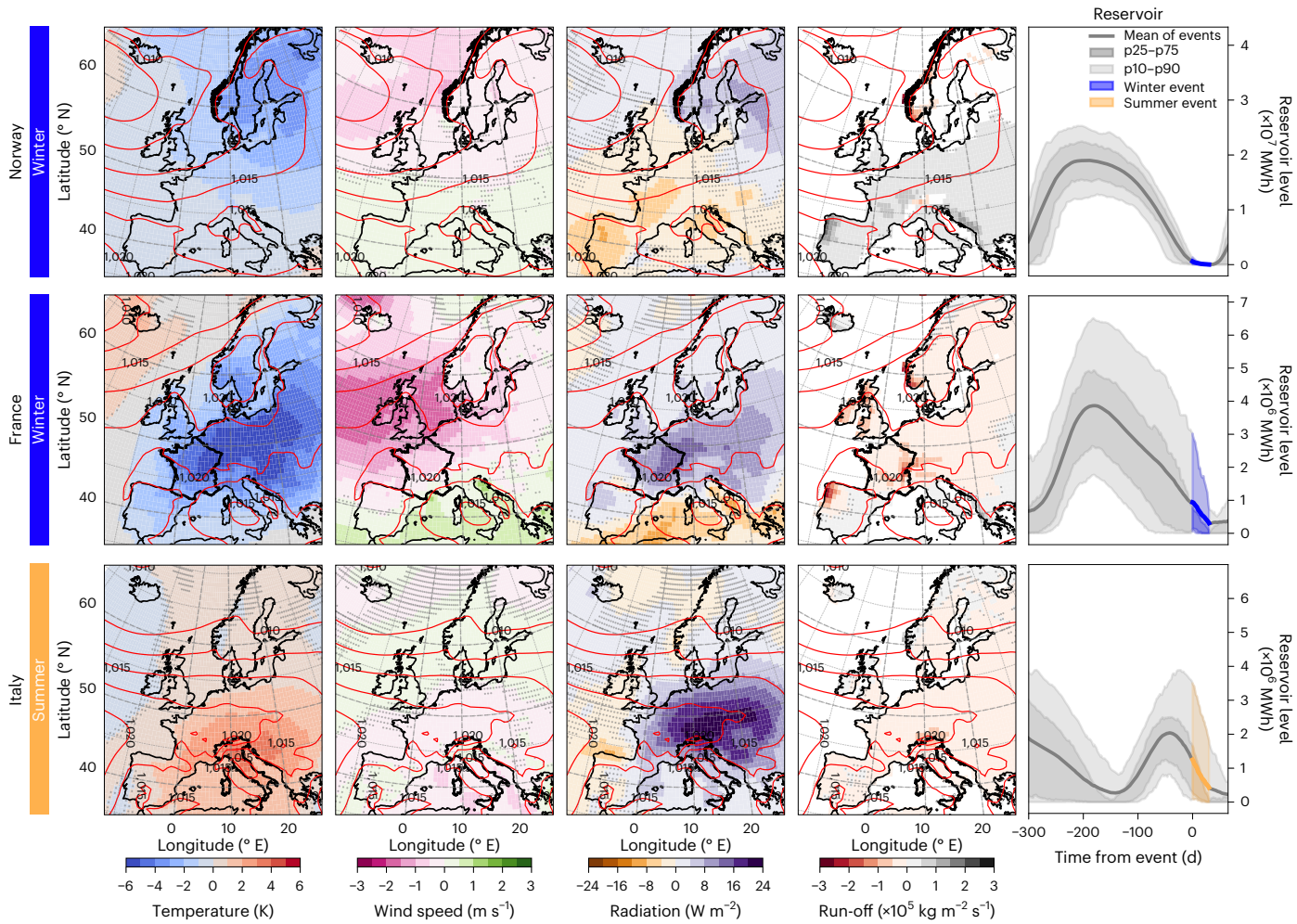

Fig. 2 | Representation of the 30-day EDWs occurring once in ten years or less. Winter EDWs for Norway (top row) and France (middle row) and summer EDWs for Italy (bottom). From left to right, the panels show anomalies of near-surface temperature, near-surface wind speed, incoming solar radiation and total run-off during the 30-day EDW period. Red contour lines indicate the mean sea level pressures during the events. On the right side, reservoir levels in the days leading up to and during the EDWs are shown. The grey areas show the 10th–90th and the 25th–75th percentiles of the hydropower reservoir levels, and the dark grey line the composite mean of all events. The blue section corresponds to the winter events, whereas the orange section corresponds to the summer events. Values that are not statistically significant (*P* value greater than 0.05) are dotted. $n = 160$.

their reservoir storage to release stored water strategically during brief periods of extreme meteorological conditions characterized by heightened demand and/or reduced renewable electricity production, mitigating short-term energy shortages. However, once the reservoir depletion exceeds a critical threshold and hydropower resources are exhausted, these systems may face long-lasting energy droughts.

In the present-day European electricity systems, energy droughts predominantly occur during the winter months (Fig. 1b–f). These can be attributed to factors such as strong heating demand and reduced wind power production (a result of the greater variability in wind speeds during the winter compared with the summer, which enhances the chance of periods with low wind[15]) and diminished reservoir storage levels. Regions in more northerly latitudes generally experience these events later in the year. This can be explained by two factors: the coldest periods occur later, and relatively large reservoir storages tend to deplete later in the year.

PEDs driven by increased cooling demands during hot summer days only occur in Spain and Italy (Fig. 1d,f)[33]. These PEDs can be intensified by reduced water availability affecting hydropower production. Because electricity demand response to high temperatures is small in Northern European countries, summer events typically transpire during early or late summer when temperatures are unseasonably low, leading to increased heating requirements. The bar plots in Fig. 1 indicate that the mean residual load during these summer events in Sweden, Norway and France is not notably higher than the average yearly residual load, meaning that in an absolute sense, summer PEDs are not extreme.

Compounding conditions during energy droughts

For the analysis of the meteorological conditions during events, we examine energy drought windows of 30 days. Figure 2 shows the meteorological conditions during these EDWs and the reservoir levels leading up to them in Norway, France and Italy (Supplementary Figs. 12–15 for other countries).

We find that prolonged high-pressure systems or ridges and atmospheric blocking are drivers of extreme low electricity production and high demand over Europe during winter[15,33]. In addition to low wind speed and low temperatures, these high-pressure systems are characterized by little to no cloud formation, potentially driving meteorological droughts[34]. This effect becomes even more pronounced during summer energy droughts, when high-pressure systems over Southern Europe form ‘heat domes’ that trap warm air in the region, causing extremely hot weather conditions. Subsidence associated with these

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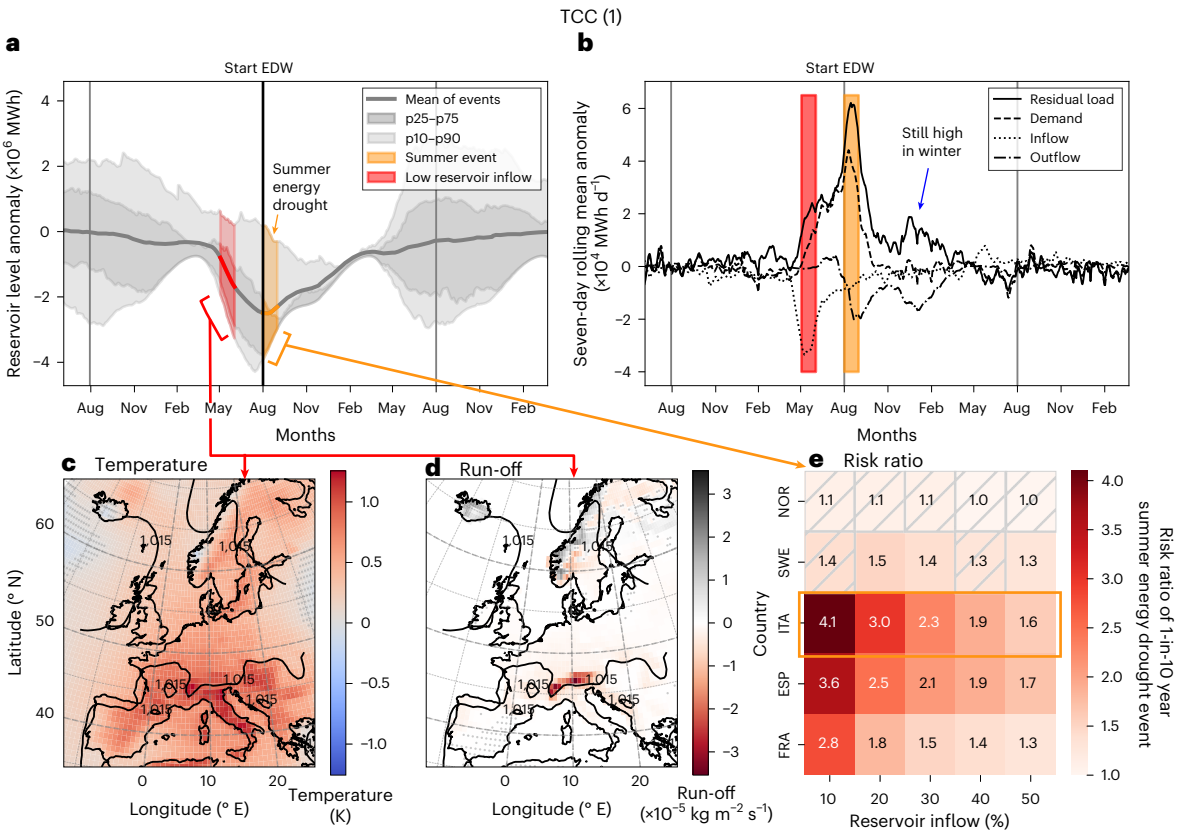

Fig. 3 | Summer energy droughts preceded by dry springs. TCC (1): the once-in-ten-years-or-less ($n = 160$) August EDW for Italy. **a**, A composite time series of the reservoir level anomalies before, during and after the events with, in red, the spring period before the summer energy drought during which a negative reservoir anomaly builds up and, in orange, the summer energy drought event. The grey areas show the 10th–90th and the 25th–75th percentiles of the hydropower reservoir levels, and the dark grey line the composite mean of all events. **b**, A plot of the temporal development of the seven-day rolling mean anomalies of residual load, demand and reservoir in- and outflow. **c**, A composite map of the near-surface temperature anomalies during the red coloured period leading up to the events with the black contour lines showing mean sea level pressures. **d**, Like **c** but for total run-off. **e**, The risk factor of a once-in-ten-years-or-less summer drought event, after low reservoir inflow in May in Norway (NOR), Sweden (SWE), Italy (ITA), Spain (ESP) and France (FRA). Values that are not statistically significant (*P* value greater than 0.05) are dotted in **c** and **d** and shaded in **e**.

systems prevent cloud formation, leading to clear skies, increased solar radiation and subsequently higher temperatures, which in turn aggravate hydrological drought conditions by strong evaporation of surface water and soil moisture[34]. Thus, during periods of extremely high demand and minimal wind production, a concurrent reduction of water inflow into reservoirs often occurs (Supplementary Fig. 16). This could cause a decline in hydropower levels that can be exacerbated by the response in hydropower dispatch. In an attempt to offset the increased demand and decreased production of wind energy, there is a rise in hydropower outflow (for example, in Italy, Fig. 2). This dual challenge—of reduced water supply occurring during heightened electricity demand—underscores the importance of managing hydropower resources effectively.

Due to the unique hydropower dispatch characteristics among the countries that we modelled, the (severity of) meteorological conditions leading to energy droughts vary considerably. In France, the meteorological anomalies during energy drought events are more extreme than in the other countries, as renewable production highly depends on non-dispatchable sources. Conversely, Sweden and Norway have a relatively large reservoir storage capacity that can produce electricity in periods of high demand. Figure 2 shows that energy droughts in Norway are linked to highly negative hydropower reservoir level anomalies, with precise meteorological conditions during the EDW, such as exceptionally low temperatures and wind speeds, playing a secondary role. In our model we find that depleted reservoir levels are a result of multiple years of imbalance between electricity demand and inflow, often caused by complex and temporally compounding processes.

Temporally compounding conditions leading to energy droughts

The climate and energy data show three types of temporally compounding condition (TCC) connected to these low reservoir levels and concurrent energy droughts: TCC (1) countries with reservoirs that are filled in the spring face vulnerability to summer energy droughts if these are preceded by dry spring conditions, resulting in low reservoir levels and a dry subsoil in the subsequent summer; TCC (2) low precipitation or run-off during reservoir-filling season compounded with persistent high-pressure systems in autumn and winter leading to winter energy droughts; and TCC (3) in certain regions (north Europe), there is an increased risk of low run-off in spring after a dry and extremely cold winter, increasing the chance of multi-year energy droughts.

Under TCC (1), the risks of a summer EDW in Italy can be up to four times greater than under normal conditions (Fig. 3e). This elevated risk occurs when a notable decline in reservoir inflow and an increase in demand during spring leaves the reservoir levels anomalously low in July and August (Fig. 3a,b, Italy serving as example). Low reservoir inflows occur when the run-off is low (Fig. 3d), which can be attributed to two factors.

First, in mountainous regions such as the Alpine area, an unusually warm winter with reduced snowfall results in a diminished snowpack by

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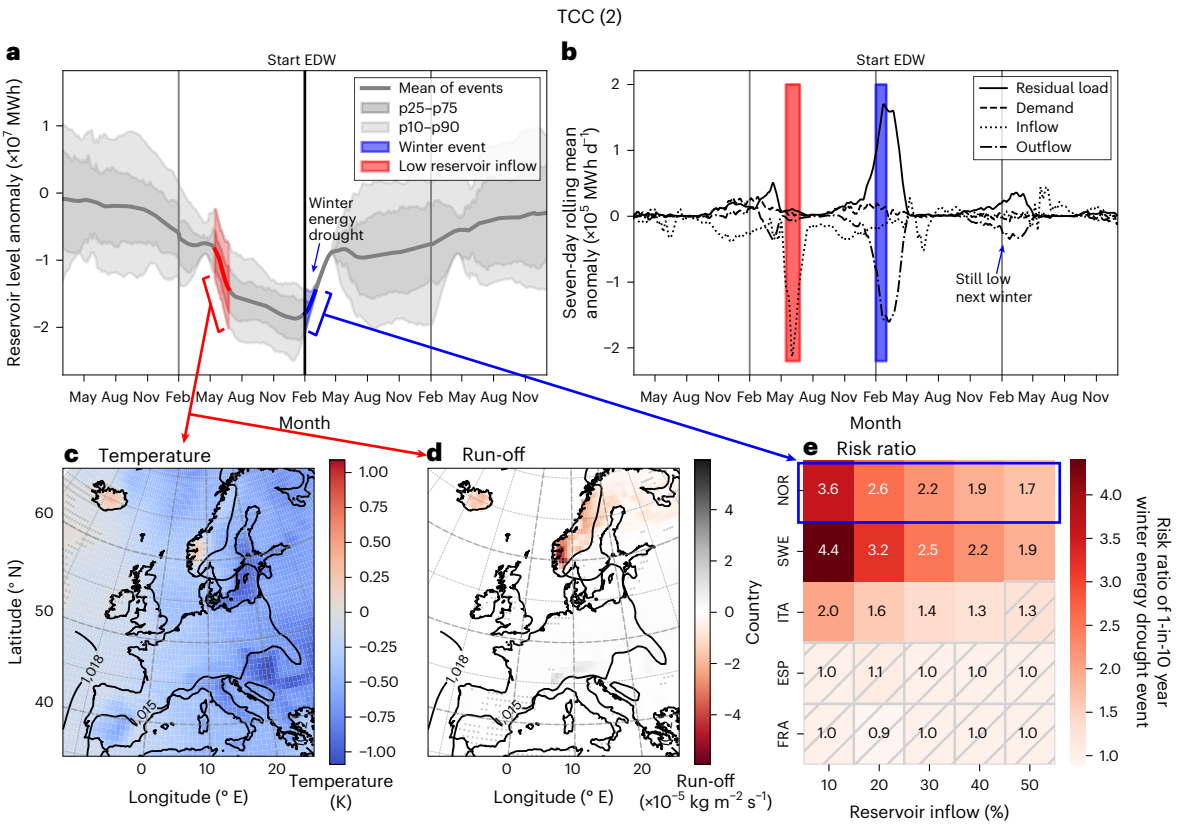

Fig. 4 | Winter energy drought preceded by low run-off during reservoir-filling season. TCC (2): the once-in-ten-years-or-less ($n = 160$) February EDWs in Norway. **a**, A composite time series of the reservoir level anomalies before, during and after the events with, in red, the spring period before the winter energy drought during which a negative reservoir anomaly builds up and, in blue, the winter energy drought event. The grey areas show the 10th–90th and the 25th–75th percentiles of the hydropower reservoir levels, and the dark grey line the composite mean of all events. **b**, A plot of the temporal development of the seven-day rolling mean anomalies of residual load, demand and reservoir in- and outflow. **c**, Composite map of the near-surface temperature anomalies during the red coloured period leading up to the events with the black contour lines showing mean sea level pressures. **d**, Like **c** but for total run-off. **e**, The risk factor of a once-in-ten-years-or-less winter drought event, after low reservoir inflow in May in Norway (NOR), Sweden (SWE), Italy (ITA), Spain (ESP) and France (FRA). Values that are not statistically significant (*P* value greater than 0.05) are dotted in **c** and **d** and shaded in **e**.

the end of winter. This reduction impacts hydropower generation later that year, as indicated in Supplementary Fig. 17. For instance, in the Po River basin of Italy, snow accumulation typically peaks between December and February. As spring arrives, rising temperatures gradually initiate the snowmelt process, releasing water into the Po River and its tributaries. Diminished snowmelt during this period leads to reduced inflow into hydropower reservoirs during spring, affecting hydropower generation in the subsequent months. The Po River drought in the year 2022 serves as a pertinent example of this phenomenon[35].

Second, during the spring season, a meteorological drought characterized by reduced precipitation due to persistent high-pressure systems also contributes to decreased inflow. Similar to the patterns observed during the summer EDW, this drought coincides with higher-than-normal temperatures (Fig. 3c).

The electricity system faces increased vulnerability to an energy drought when not only supply is low (depleted reservoirs) but concurrently demand is high (cooling demand during exceptional heat). Dry winter and spring conditions not only leave reservoirs depleted but also result in dry subsoil. Note that the risk of exceptionally hot summers is heightened in these cases, due to this dry subsoil and land–atmosphere feedback mechanisms[36,37].

So, although by design, the energy drought window lasts 30 days, Fig. 3b demonstrates how the compounding of these conditions can lead to extended periods of high residual loads. Besides the elevated demand during a warmer-than-normal spring, at the end of a summer, temperatures may return to normal, but the reservoirs have been depleted, resulting in prolonged periods of high residual loads while entering the subsequent winter.

Under TCC (2) (Fig. 4), the combination of low filling of reservoirs in spring and subsequent persisting high-pressure systems in the following winter set the stage for winter energy droughts. The risk of a winter energy drought following such conditions is 3.6 times higher in Norway and up to 4.4 times higher in the other countries (Fig. 4e). Figure 4 shows that for Norway, as in TCC (1), strongly reduced run-off in May leads to reduced reservoir levels. As winter approaches, prolonged high-pressure systems can reduce precipitation and thereby oppose the replenishment of these reservoirs. Together with increased heating demands associated with these weather systems (Fig. 4b), residual loads can be increased for long periods of time leading up to winter events. Importantly, we find that the inflow into the reservoirs remains low during subsequent filling seasons after the occurrence of winter EDWs, which works against the recovery and robustness of the entire system.

In TCC (3) (Fig. 5, Sweden serving as example), winter energy droughts are followed by lower-than-usual reservoir inflows during the subsequent spring season. Precipitation, evaporation and snowmelt after winter EDWs show little to no deviation during the filling season when compared with years without such events, but the run-off (Fig. 5d) and consequently reservoir inflow (Fig. 5b) is very low. We predominantly observe this in Sweden and Norway, particularly when winter EDWs are preceded by dry springs and summers (as detailed in TCC (2)), rendering the electricity system vulnerable, but also leaving

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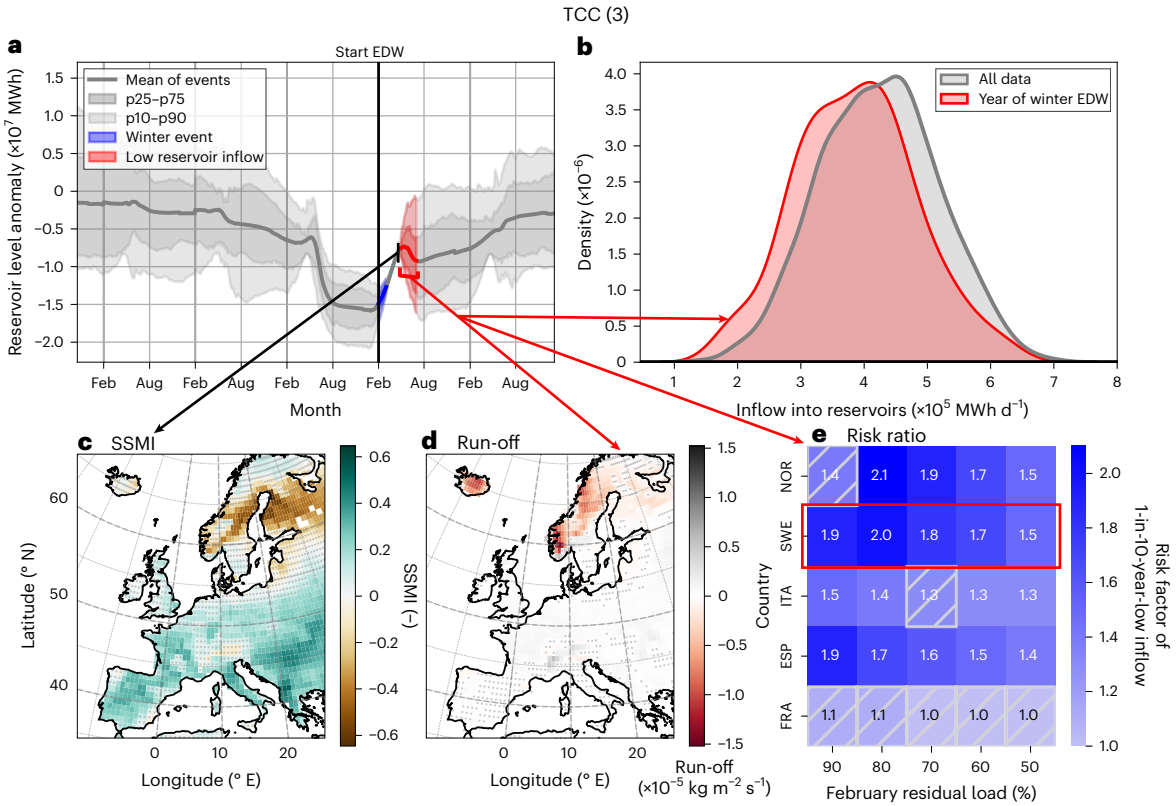

Fig. 5 | Low inflow into reservoirs the season after an energy drought. TCC (3): the once-in-ten-years-or-less ($n = 160$) February EDW for Sweden and the reservoir-filling season afterwards. **a**, A composite time series of the reservoir level anomalies before, during and after the events with, in red, the spring period after the winter energy drought during which a negative reservoir anomaly builds up and, in blue, the winter energy drought event. The grey areas show the 10th–90th and the 25th–75th percentiles of the hydropower reservoir levels, and the dark grey line the composite mean of all events. **b**, The distribution of inflow during the red period for all 1,600 years and for the years with a preceding energy drought in winter. **c**, A composite map of the Standardized Soil Moisture Index (SSMI) at the start of May. **d**, A composite map of total run-off anomalies during the red coloured period after the drought events. **e**, The risk factor of a once-in-ten-years-or-less low inflow in spring after high residual loads in February in Norway (NOR), Sweden (SWE), Italy (ITA), Spain (ESP) and France (FRA). Values that are not statistically significant (P value greater than 0.05) are dotted in **c** and **d** and shaded in **e**.

the soil dry. This soil moisture drought is exacerbated during the winter EDWs, during which episodes of extreme cold coincide with reduced run-off, as can be observed in Fig. 2.

This dry subsoil (Fig. 5c) is the main cause of continued low run-off in spring. Precipitation following an energy drought initially infiltrates the soil to replenish the moisture deficit before eventually moving via run-off into streamflow and ultimately flowing into the hydropower reservoirs. This particular phenomenon nearly doubles the risk of experiencing extremely low inflow into the reservoirs, opposing the chance for the electricity system to rebound. Figure 5a shows that up to three years after the occurrence of a winter EDW, the mean hydropower reservoirs levels remain anomalously low.

Prolonged, cumulative energy droughts

In three out of the five studied countries, we found that as a result of these compounding events, energy droughts are rarely isolated events but rather a consequence of feedback mechanisms that tend to prolong periods of energy scarcity (Fig. 6a) and can turn what might have been a short-term drought into prolonged high residual loads.

The sequence of the three temporally compounding conditions creates a cyclical pattern, feeding into each other. Typically, low reservoir inflows in the spring season (April, May, June) are caused by a thinner snowpack and/or reduced precipitation. These conditions also lead to relatively dry subsoils, which in turn increase the likelihood of extreme heat events during the summer months. The combination of diminished inflow and increased demand for cooling can lead to an energy shortage in the summer (July, August), TCC (1). If the low inflows persist, reduced reservoir levels by winter may trigger an energy drought (December, January, February, March), TCC (2). Furthermore, in Norway and Sweden, there is a notable risk of decreased spring run-off (May, June) due to anomalously high residual loads. This is attributed to the low soil moisture conditions originating from previous periods, TCC (3), further enhancing the feedback loop.

At least partially due to this feedback loop, the sets of ten-year climate model simulations that contain one or more energy droughts tend to experience a progressive increase in the residual load anomaly over time (Fig. 6b,c), suggesting that decadal climate variability plays a role in determining the occurrence of prolonged energy droughts in Europe.

Discussion and conclusions

We have highlighted the complexity of the impact of meteorological conditions on fluctuating electricity demands in the face of diminished renewable energy production. Within this context, we have identified three distinct types of temporally compounding conditions. Furthermore, the results suggest that these three TCCs may be an intrinsic part of a cyclical pattern that plays a role in decadal variability in residual load anomalies.

This exploration suggests that relying solely on multiple individual weather years to assess system robustness, as is traditionally done, is inadequate to capture the increased risk and intricate nature associated with multi-year energy droughts. This seems particularly relevant for hydroelectrical systems, for which, due to the temporally compounding nature of drought events, evaluating consecutive multi-year data

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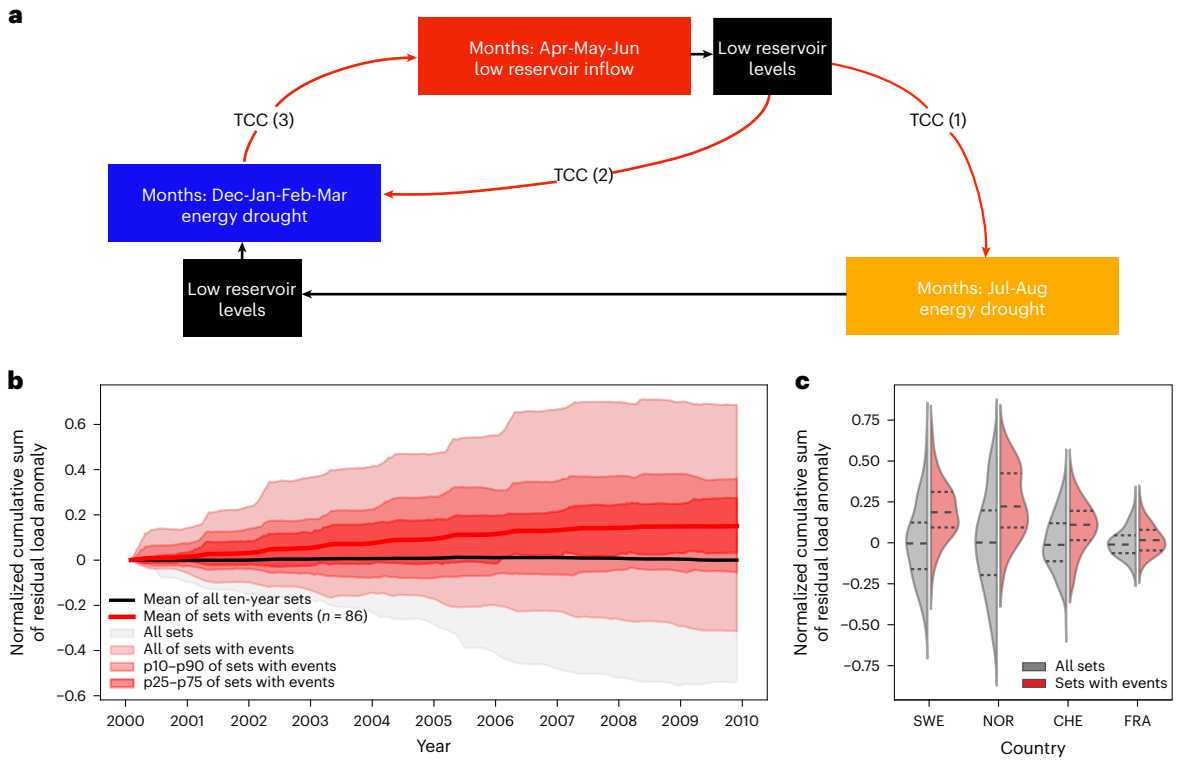

Fig. 6 | The reinforcing cycle of multi-year energy droughts. a, A schematic overview of the energy drought cycle. (Red arrows implicate increased risk as a result of temporally compounding conditions, black arrows a causal link.) **b**, The cumulative sum of residual load anomalies normalized over yearly demand values with in red the sets of the 160 climate model simulations that include energy drought events ($n = 86$) and in grey all 160 simulations. **c**, Violin plots of the normalized residual load anomalies during the last time step of the simulations for three countries. The dashed lines within each violin represent quartiles, with the middle line indicating the median.

becomes essential for a comprehensive understanding and planning for extreme scenarios.

We found that for Sweden and Norway, managing their hydropower reservoirs during winter energy droughts is important as the risks of periods of low reservoir inflow increase after energy droughts, rendering the system's recovery more challenging. On the basis of the observation that meteorological conditions leading to electricity system stress often align with periods of anomalously high solar radiation, southern European countries with large hydropower capacities could potentially mitigate summer energy droughts by the investment in additional solar panels.

We based these analyses on a dataset from a single global climate model that does not incorporate glacier dynamics and melt. Although this dataset has been assessed against reanalysis and pseudo-observational data for different regions, it has not been bias corrected. And although the assessment of the dataset indicates plausibility, an absence of observational datasets for run-off poses challenges in validation, especially of extremes. Future research should evaluate the consistency of our findings across multiple climate models to ensure robustness.

We employed streamlined models to manage extensive climate datasets generated with a physical climate model, enabling the analysis of extreme conditions without resorting to statistical extrapolation. This approach facilitates the identification of compounding conditions, while preserving the physical integrity of climate relationships but imposes certain limitations. By not including transmission and grid interconnectivity, other dispatchable sources and unit commitment, our analysis might overlook how network dynamics could mitigate or exacerbate risk in response to concurrent high residual loads in adjacent countries. To focus on isolating the impact of weather on energy demand peaks we modelled demand as a function of temperature, omitting economic and demographic influences. Similarly, by modelling hydroelectric systems merely as inflow and outflow at the national level, our model does not account for human responses to droughts, such as upstream flow regulation or changes in reservoir management, which might mitigate impacts. Finally, we have focused on variability in the three main types of renewable production in Europe, but there are additional weather dependencies that might aggravate the effects of meteorological conditions on the energy system. For example, during periods of reduced discharge (leading to low reservoir inflows) there might be concurrent shortfall in freshwater used for cooling by thermal power plants[38,39], and weather extremes such as storms pose an additional risk of infrastructure being damaged[40,41].

So although this model gives good insight in general sensitivity to meteorological conditions, a number of these simplifications lead to poor reproduction of the actual situation (such as daily load in Italy or hydropower production in Switzerland) in the energy system. Overall, our results should therefore be interpreted as an exploration of energy system vulnerabilities, providing a starting point for further research rather than a definitive forecast of system-wide resilience. We recommend using the findings as a basis for more detailed analyses including but not limited to water management and grid dynamics models.

These findings accentuate the need for strategic planning in the face of energy droughts, which may become more pronounced as climate change progresses and as the European Union shifts towards increased reliance on renewable energy sources to meet its Green Deal objectives. It is therefore essential to align energy system upgrades with research into climate variability to ensure that renewable energy goals are met without compromising system reliability.

Additional next research phases would be to examine how climate change and the energy transition affect long-lasting energy droughts. Increased renewable energy capacities and more intense/frequent

weather extremes may lead to even greater challenges in future energy management. The interplay between a changing climate and electricity system adaptations needs analyses to develop strategies mitigating the impacts of extended energy shortages, crucial for shaping sustainable energy policies in an era of climate uncertainty.

Methods

Meteorological data

We use the daily outputs from the KNMI Large ENsemble TIme Slice (KNMI-LENTIS) dataset[31], a large climate ensemble composed of 160 sets of ten-year physical climate model simulations of present-day climate (the time slice 2000–2009; Supplementary Note 2 and Supplementary Fig. 2). The KNMI-LENTIS ensemble was executed with EC-Earth3, a fully coupled, state-of-the-art global climate model that includes representations of the physical processes in the atmosphere, oceans, land and sea ice[42].

This time slice ensemble methodology is especially suitable for investigating extreme events. The 1,600 years of data allow for the examination of 160 extreme events that occur once in ten years or less. Moreover, the sequence of ten continuous weather years in each simulation makes it possible to study temporally compounding and multi-year events.

We did not bias correct the KNMI-LENTIS data before application in the energy modelling framework. In the context of this multivariate research, bias correction of climate data may unintentionally distort the physical relationships between different variables and the seasons, potentially leading to inaccurate interpretations of climate interactions and dynamics.

However, to demonstrate the reliability and variance of the KNMI-LENTIS data, we benchmarked its climatological mean and variance of meteorological variables for each country against the climatology of three reanalyses datasets (ERA5[43], MERRA-2[44] and JRA-55[45]) for the years 1990–2019. For a more complete analysis of the run-off data, we included two pseudo-observational datasets (GRUN[46] and LORA[47]); Supplementary Note 3 with Supplementary Figs. 3–7. The KNMI-LENTIS data exhibit a warm bias in the Southern European regions in summer, implicating that actual summer demands are somewhat lower than modelled. Overall, KNMI-LENTIS wind speeds remain within the margin of difference between the reanalysis data but show a small positive wind speed bias in north Europe.

For run-off, the comparative analysis showed notable variability across the models, with the KNMI-LENTIS dataset indicating an earlier and higher run-off peak in Norway and Sweden with much larger variance and slightly lower run-off in the beginning of the year in France, Italy and Spain compared with the reanalysis and pseudo-observational datasets. Direct large-scale run-off measurements are not feasible, and there are notable differences between the pseudo-observational datasets themselves. Consequently, whereas the model provides valuable insights and is a potential representation of present-day climate, there remains a level of uncertainty.

Electricity production and demand modelling framework

The electricity modelling framework includes gridded production modules for offshore wind, onshore wind, solar PV, run-of-river and hydropower reservoir inflow. It also contains a hydropower dispatch module and an electricity demand module, both operating at the European country level[48]. A flowchart of the modelling framework is included in Supplementary Note 4 (Supplementary Fig. 8).

Solar PV and wind production

For wind and solar PV power modules, the inputs include 10-m wind speed (m s^{-1}), near-surface temperature (°C), near-surface maximum daily temperature (°C) and solar irradiance (kW m^{-2}). We calculate wind power by extrapolating 10-m wind speeds to hub height in both onshore and offshore scenarios using the power law equation. We then estimate the power generation through a cubic power curve with onshore and offshore specific cut-in, rated and cut-out wind speeds, neglecting downtime and efficiency losses.

We base solar PV power on solar radiation and temperature-dependent cell efficiency, which we determine using near-surface temperature, maximum daily temperature and wind speed. We assume that all PV modules are horizontally oriented and functional during all daylight hours.

Demand curves

For the demand model, we use a logistic smooth transmission regression approach to fit daily load data from the ENTSO-E Transparency Data Platform[49] to national mean, population-weighted daily near-surface temperatures (°C) from ERA5 reanalysis data[43]. The objective is to identify meteorological drivers that could potentially stress the energy system. Thus, our emphasis is on assessing how weather conditions impact energy demand, not on producing exact demand forecasts, corresponding with other work in the energy meteorology field[15,50].We differentiate between weekdays and weekends to account for variations in heating and cooling behaviours. Cultural and social-economic influences such as holidays are neglected. To avoid unreliable extrapolation for extreme temperature scenarios, we capped the linear regime at the maximum daily load to date. Supplementary Note 4 includes figures on the demand fit and validation data (Supplementary Figs. 10 and 11 respectively).

Routing run-off

To simulate water discharge, the run-off data are processed through a routing scheme that integrates flow direction and accumulation across a grid with a resolution of 0.5° × 0.5°. This routing begins at the highest points in a basin and proceeds downstream, aggregating run-off as it goes. Discharge at each grid cell is calculated while considering a recession coefficient that accounts for the rate of flow reduction over time. The model does not incorporate delays from lake accumulation, human water use or water management practices such as dams.

Inflow into hydropower reservoirs

The potential energy available for hydropower in each grid cell is estimated based on the discharge, water density, gravitational acceleration, hydraulic head and an efficiency factor for the hydropower plant. To accommodate the large grid cell size, we adjust the discharge in each grid cell according to its installed hydropower capacity, using a fraction-of-discharge factor derived from the national mean annual capacity factor (derived from ENTSO-E data[49]) and average yearly discharge (of the KNMI-LENTIS dataset). The resulting energy from reservoir inflow is then aggregated on a national level to estimate the country's total potential hydropower inflow.

Modelling run-of-river production

For the run-of-river hydropower production module, the utilizable discharge is dictated by the installed capacity, with an assumed exceedance probability of 25%. This exceedance reflects the plant's design limitations for harnessing discharge above its capacity. We set a discharge fraction to correct for the utilizable portion in each grid cell, based on a comparison between the installed capacity's maximum discharge and the 75th percentile discharge in an average year (based on the KNMI-LENTIS dataset). Without considering upstream reservoir storage or downtime for maintenance, the run-of-river electricity production is estimated from available power based on natural discharge over each time step.

The hydropower dispatch model

The hydropower dispatch is modelled at the national level using a linear optimization approach that aims to minimize the difference between demand and the sum of all renewable energy production over a year,

directing the solution to following the load curves. This approach aims not only to maximize the utilization of available hydropower throughout the year but also to prioritize its use during periods of high residual loads. The rationale behind this strategy is the assumption that energy prices tend to increase during times of high residual load. Supplementary Fig. 9 shows that although this is a simplification of a full energy system, it reasonably approaches the national hydropower dispatch cycle, making it suitable for a preliminary vulnerability assessment. The optimization utilizes the 160 ten-year datasets, progressing in 28-day intervals for the course of a year. To prevent perfect foresight, the initial 42 days of each optimization window employ specific KNMI-LENTIS data, whereas the remaining period uses an ensemble mean.

To determine hydropower fill levels at the beginning of each year, we first run the model with a perfect foresight optimization and from these results make a randomized selection of the fill levels on 1 January for the start of the rolling window optimization. To capture system variability, we exclude the first and last months of the ten-year time series, which are constrained by the dispatch model.

A more detailed description of all the modules within the modelling framework, along with a validation of how it performs on the selection of energy droughts with the use of ERA5 data, are available in ref. 30.

Installed capacities

We conducted the study using gridded files that reflect Europe's 2020 installed renewable capacities. We took the 2020 locations of solar photovoltaic and onshore windfarms from Dunnet et al.[51] and remapped them to the KNMI-LENTIS grid. Dunnet et. al developed an algorithm to extract the locations of solar photovoltaic and onshore wind farms from OpenStreetMap. EMODnet[52] provides vector data of installed offshore wind capacities, which we converted to gridded datasets. We sourced the locations of run-of-river and reservoir hydropower plants from the Joint Research Centre (JRC) Hydropower database[53]. Recognizing the gaps in the head data (the height through which water falls to generate power), we employed a methodical approach to estimate these missing values. We calculated the elevation difference between the highest and lowest geographical points surrounding each hydropower dam's vicinity. This elevation difference was then statistically correlated with the head data already present in the JRC database. The established correlation was subsequently applied to infer the missing head data across the dataset. A more elaborate description on the creation of the installed capacities can be found in ref. 30.

Selection of drought events

Our analysis targets energy drought events that have a return period of ten years or longer. To this end, we selected the top 160 events from our 1,600-year database. PEDs are defined as periods where the residual load exceeds the 97th percentile threshold for several consecutive days. A PED concludes when the load falls below this threshold for more than three consecutive days. The PEDs are ranked in descending order of duration, focusing on the 160 longest events to identify the most extreme cases. Differentiation between summer and winter events is made, with each category benchmarked against their specific 97th percentile threshold.

To evaluate the effects of meteorological compounding conditions in these events, we analysed same-length events in the form of EDWs that span 30 days or are confined to a specific month. This selected duration aligns with the approximate average length of PEDs as observed across the countries included in our analysis. The 160 EDWs with the highest average residual loads (and without overlap) are selected to identify the most impactful energy droughts.

Risk ratios

We quantify and compare relative risks of energy droughts under distinct temporally compounding meteorological conditions using calculated risk ratios (RRs): the ratio of the probability of an energy drought occurring after certain meteorological conditions (for example, low inflow into hydropower reservoirs in spring) to the climatological probability. Risk ratios[54] are commonly used in compound climate studies as they provide a quantitative measure of the association between exposure to a certain climate event (or events) and the likelihood of a specific outcome (in this cause energy droughts). We assessed the statistical significance and precision of the estimated RR by computing its 95% confidence interval (CI) according to $CI = \exp(\log RR \pm Z \times SE(\log RR))$, where RR is the risk ratio, SE is the standard error of the log-transformed risk ratio and Z represents the standard normal deviate (1.96 for a 95% CI). Risk ratios for which the 95% CI does not include 1 are considered statistically significant.

Standardized Soil Moisture Index

We analyse soil moisture variability after winter energy droughts with the Standardized Soil Moisture Index (SSMI)[55]. The SSMI uses the z-score methodology to quantify soil moisture deviations in terms of standard deviations from the mean and is calculated using the formula: $SSMI = (mrso - \mu)/s.d.$, where mrso represents the soil moisture value for a given day, μ is the mean soil moisture over the 1,600-year period on that day of year and s.d. is the standard deviation of soil moisture during that day of year.

Data availability

All data supporting the findings of this study and required to reproduce the figures are accessible at https://github.com/L-vdM/Temporally-compounding-energy-droughts-figures. The full energy datasets generated and analysed during the current study are available via Zenodo at https://doi.org/10.5281/zenodo.12634375 (ref. 56).

Code availability

The code used to select extremes and calculate risk ratios is available via GitHub at https://github.com/L-vdM/energydroughts-Europe.

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Author contributions

L.v.d.M., K.v.d.W., R.M.J.B., P.W.G.-L. and R.B. conceived and designed the analysis. L.v.d.M. performed simulations, data analysis and drafted the paper. K.v.d.W., R.M.J.B., P.W.G.-L. and R.B. helped with the interpretation and analysis of the results, reviewed the paper and supervised the work.

Competing interests

The authors declare no competing interests.

Additional information

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